

MAT 115E Introduction to Programming Language

Lab-8/ CRN : 23492

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1 Question 1

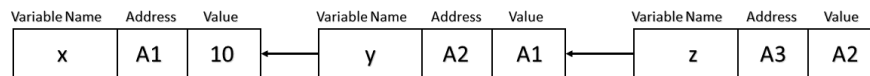


Figure 1: Memory state

Write a C program that performs the following, based on the memory state described above.

- Design a C program that implements the memory state.
- Set the value of variable x to 20 using the variable y.
- Set the value of variable x to 30 using the variable z.

2 Question 2

Write a C program that swaps two pointers using an additional pointer as illustrated in the following figures.

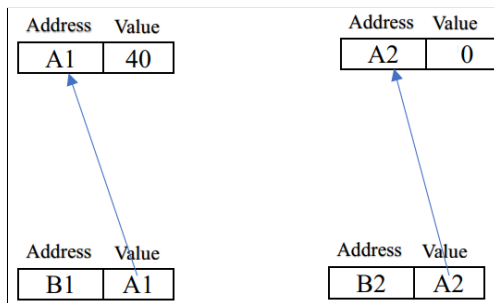


Figure 2: Status of the pointers before swapping

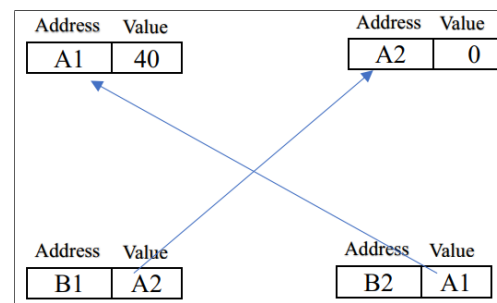


Figure 3: Status of the pointers after swapping

3 Question 3

Definition: A lower triangular matrix is a square matrix in which all elements above the main diagonal are zero. Mathematically,

For a matrix $A = [a_{ij}]$, if

$$i < j \Rightarrow a_{ij} = 0$$

then A is a lower triangular matrix.

Write a C program that reads a square matrix and checks whether it is lower triangular or not.